Reg. No									

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001

EMT512 – MECHANICS - II

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. Define Kinetic energy.
2. Define horizontal range.
3. Define direct impact.
4. Write down the definition for force.
5. Write down the M.I of solid sphere of radius a.

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. Find the power of the pump which lifts 3000 litres of water per minute from a well 10 metres deep and projects it with a velocity of 16 m/sec (1 litre water is of mass 1 kg)
7. If v_1 and v_2 are the velocities of a projectile at the ends of a focal chord of its path and v, the horizontal component of its velocity then, show that $\frac{1}{v_1^2} + \frac{1}{v_2^2} = \frac{1}{v^2}$.
8. A gun is rigidly mounted on a carriage on a smooth horizontal palne. The total mass of the carriage and the gun is *M*. If the gun is

elevated at an angle α to the horizon, show that when a shot of mass *m* is fired, it leaves the gun in a direction at an angle θ to the horizontal given by $tan\,\theta = \frac{M+m}{M} tan\,\alpha$

9. Show that the path of a point *P* whose velocity is such that its components in a fixed direction and in the direction perpendicular to the line joining *P* to a fixed point *O* are respectively the constants *u* and *v*, is a conic with a focus at *O* and eccentricity u/v.

10. Find the M.I of a square about its diagonal of length *l*.

SECTION - C (5X10=50)

Answer **ALL** Questions

11. a) A car of mass1tonne attains a maximum speed of 45 km.p.h when freely running down an incline of 1 in 10. What power must its engine develop to take it up an incline of 1 in 20 with the same speed if in both the cases the resistance is the same?

(or)

b) Show that the resultant motion of simple harmonic motions of same period along two perpendicular lines is along an ellipse.

12. a) A particle is projected with A velocity whose horizontal and vertical components are *U* and *V*, If P_1 and P_2 are the points on its trajectory at the same height *h* above the point of projection, show that $P_1P_2 = \frac{2U}{g}\sqrt{v^2 - 2gh}$

(or)

b) Show that the path of a projectile is a parabola.

13. a) A shell of mass *m* is moving with velocity *v*, An internal explosion generates an energy *E* and breaks the shell into two portions whose masses are in the ratio a : b. They continue to

move in the original line of motion then show that, their velocities after explosion are $v + \sqrt{\frac{2bE}{am}}$, $v - \sqrt{\frac{2aE}{bm}}$

(or)

b) A ball dropped from a height *h* on a horizontal plane bounces up and down. If the coefficient of restitution is *e*, then prove that

i) The whole distance *H* covered before it comes to rest is $h\frac{1+e^2}{1-e^2}$

ii) The total time *T* taken is $\frac{1+e}{1-e}\sqrt{\frac{2h}{g}}$

14. a) Find the equation of the central orbit in polar co-ordinates.

(or)

b) A particle moves with a central acceleration μr^{-7} and starts from an apse at a distance α with a velocity equal to the velocity which would be acquired by the particle travelling from rest at infinity to the apse then show that the equation of its orbit is $r^2 = a^2\cos 2\theta$.

15. a) Show that the M.I. of a rectangular lamina of mass *M* and sides 2*a* and 2*b* about a diagonal is $M\frac{2a^2b^2}{3(a^2+b^2)}$

(or)

b) Find the moment of interia of parabolic lamina.

****